$$Y_d^{2\,i1\_i2\_}\to$$

$$-y_d^{i1i2}+\frac{1}{16\pi^2}\left(\frac{1}{48}\left(y_d^{i1i2}\left(g_1^2-9g_2^2-96g_3^2\right)-6y_d^{pi2}\left(3\overline{y_d}^{pr}y_d^{i1r}+\overline{y_u}^{pr}y_u^{i1r}\right)\right)+\frac{1}{9}g_1^2y_d^{i1i2}\right.$$

$$LF_{1,1,0}[m_1,m_{\tilde{d}}^{i2}]-\frac{1}{18}g_1^2y_d^{i1i2}LF_{2,1,-1}[m_1,m_{\tilde{d}}^{i2}]+\frac{1}{36}g_1^2y_d^{i1i2}LF_{1,1,0}[m_1,m_{\tilde{q}}^{i1}]-$$

$$\frac{1}{72}g_1^2y_d^{i1i2}LF_{2,1,-1}[m_1,m_{\tilde{q}}^{i1}]+\frac{3}{4}g_2^2y_d^{i1i2}LF_{1,1,0}[m_2,m_{\tilde{q}}^{i1}]-$$

$$\frac{3}{8}g_2^2y_d^{i1i2}LF_{2,1,-1}[m_2,m_{\tilde{q}}^{i1}]+\frac{4}{3}g_3^2y_d^{i1i2}LF_{1,1,0}[m_3,m_{\tilde{d}}^{i2}]-$$

$$\frac{2}{3}g_3^2y_d^{i1i2}LF_{2,1,-1}[m_3,m_{\tilde{d}}^{i2}]+\frac{4}{3}g_3^2y_d^{i1i2}LF_{1,1,0}[m_3,m_{\tilde{q}}^{i1}]-$$

$$\frac{2}{3}g_3^2y_d^{i1i2}LF_{2,1,-1}[m_3,m_{\tilde{q}}^{i1}]+\frac{1}{2}\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{1,1,0}[m_{\tilde{d}}^{r},\tilde{\mu}]-$$

$$\frac{1}{2}y_d^{i1i2}\overline{a_e}^{pr}a_e^{pr}LF_{2,1,0}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]+\frac{1}{2}y_d^{i1i2}\overline{a_e}^{pr}a_e^{pr}LF_{3,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]-$$

$$\frac{1}{2}y_d^{i1i2}\left(C_{\phi2\phi1}\tilde{\mu}\overline{y_e}^{pr}a_e^{pr}+\overline{a_e}^{pr}\left(2C_{\phi2^2}a_e^{pr}+\overline{C_{\phi2\phi1}}\tilde{\mu}y_e^{pr}\right)\right)LF_{3,1,0}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]+$$

$$\frac{3}{2}y_d^{i1i2}\left(C_{\phi2\phi1}\tilde{\mu}\overline{y_e}^{pr}a_e^{pr}+\overline{a_e}^{pr}\left(2C_{\phi2^2}a_e^{pr}+\overline{C_{\phi2\phi1}}\tilde{\mu}y_e^{pr}\right)\right)LF_{4,1,-1}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]-$$

$$y_d^{i1i2}\left(C_{\phi2\phi1}\tilde{\mu}\overline{y_e}^{pr}a_e^{pr}+\overline{a_e}^{pr}\left(2C_{\phi2^2}a_e^{pr}+\overline{C_{\phi2\phi1}}\tilde{\mu}y_e^{pr}\right)\right)LF_{5,1,-2}[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}]-$$

$$\frac{3}{2}y_d^{i1i2}\overline{a_d}^{pr}a_d^{pr}LF_{2,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]+\frac{3}{2}y_d^{i1i2}\overline{a_d}^{pr}a_d^{pr}LF_{3,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]-$$

$$\frac{3}{2}y_d^{i1i2}\left(C_{\phi2\phi1}\tilde{\mu}\overline{y_d}^{pr}a_d^{pr}+\overline{a_d}^{pr}\left(2C_{\phi2^2}a_d^{pr}+\overline{C_{\phi2\phi1}}\tilde{\mu}y_d^{pr}\right)\right)LF_{3,1,0}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]+$$

$$\frac{9}{2}y_d^{i1i2}\left(C_{\phi2\phi1}\tilde{\mu}\overline{y_d}^{pr}a_d^{pr}+\overline{a_d}^{pr}\left(2C_{\phi2^2}a_d^{pr}+\overline{C_{\phi2\phi1}}\tilde{\mu}y_d^{pr}\right)\right)LF_{4,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]-$$

$$3y_d^{i1i2}\left(C_{\phi2\phi1}\tilde{\mu}\overline{y_d}^{pr}a_d^{pr}+\overline{a_d}^{pr}\left(2C_{\phi2^2}a_d^{pr}+\overline{C_{\phi2\phi1}}\tilde{\mu}y_d^{pr}\right)\right)LF_{5,1,-2}[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}]+$$

$$\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{1,1,0}[m_{\tilde{q}}^{p},\tilde{\mu}]-\frac{3}{2}\tilde{\mu}^2y_d^{i1i2}\overline{y_u}^{pr}y_u^{pr}LF_{2,1,0}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]+$$

$$\frac{3}{2}\tilde{\mu}^2y_d^{i1i2}\overline{y_u}^{pr}y_u^{pr}LF_{3,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]-$$

$$\frac{3}{2}\tilde{\mu}y_d^{i1i2}\left(\overline{C_{\phi2\phi1}}y_u^{pr}\overline{a_u}^{pr}+\overline{y_u}^{pr}\left(C_{\phi2\phi1}a_u^{pr}+2C_{\phi2^2}\tilde{\mu}y_u^{pr}\right)\right)LF_{3,1,0}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]+$$

$$\frac{9}{2}\tilde{\mu}y_d^{i1i2}\left(\overline{C_{\phi2\phi1}}y_u^{pr}\overline{a_u}^{pr}+\overline{y_u}^{pr}\left(C_{\phi2\phi1}a_u^{pr}+2C_{\phi2^2}\tilde{\mu}y_u^{pr}\right)\right)LF_{4,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]-$$

$$3\tilde{\mu}y_d^{i1i2}\left(\overline{C_{\phi2\phi1}}y_u^{pr}\overline{a_u}^{pr}+\overline{y_u}^{pr}\left(C_{\phi2\phi1}a_u^{pr}+2C_{\phi2^2}\tilde{\mu}y_u^{pr}\right)\right)LF_{5,1,-2}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}]+$$

$$\frac{1}{2}y_d^{pi2}\overline{y_u}^{pr}y_u^{i1r}LF_{1,1,0}[m_{\tilde{u}}^{r},\tilde{\mu}]+\frac{3}{4}g_1^2y_d^{i1i2}LF_{2,1,-1}[\tilde{\mu},m_1]-\frac{1}{2}g_1^2y_d^{i1i2}LF_{3,1,-2}[\tilde{\mu},m_1]+$$

$$2C_{\phi2^2}g_1^2y_d^{i1i2}LF_{3,1,-1}[\tilde{\mu},m_1]-\frac{1}{2}m_1\tilde{\mu}g_1^2y_d^{i1i2}\left(\overline{C_{\phi2\phi1}}+C_{\phi2\phi1}\right)LF_{3,1,0}[\tilde{\mu},m_1]-$$

$$4C_{\phi2^2}g_1^2y_d^{i1i2}LF_{4,1,-2}[\tilde{\mu},m_1]+\frac{3}{2}m_1\tilde{\mu}g_1^2y_d^{i1i2}\left(\overline{C_{\phi2\phi1}}+C_{\phi2\phi1}\right)LF_{4,1,-1}[\tilde{\mu},m_1]+$$

$$2C_{\phi2^2}g_1^2y_d^{i1i2}LF_{5,1,-3}[\tilde{\mu},m_1]-m_1\tilde{\mu}g_1^2y_d^{i1i2}\left(\overline{C_{\phi2\phi1}}+C_{\phi2\phi1}\right)LF_{5,1,-2}[\tilde{\mu},m_1]+$$

$$\frac{9}{4}g_2^2y_d^{i1i2}LF_{2,1,-1}[\tilde{\mu},m_2]-\frac{3}{2}g_2^2y_d^{i1i2}LF_{3,1,-2}[\tilde{\mu},m_2]+$$

$$6C_{\phi2^2}g_2^2y_d^{i1i2}LF_{3,1,-1}[\tilde{\mu},m_2]-\frac{3}{2}m_2\tilde{\mu}g_2^2y_d^{i1i2}\left(\overline{C_{\phi2\phi1}}+C_{\phi2\phi1}\right)LF_{3,1,0}[\tilde{\mu},m_2]-$$

$$12C_{\phi2^2}g_2^2y_d^{i1i2}LF_{4,1,-2}[\tilde{\mu},m_2]+\frac{9}{2}m_2\tilde{\mu}g_2^2y_d^{i1i2}\left(\overline{C_{\phi2\phi1}}+C_{\phi2\phi1}\right)LF_{4,1,-1}[\tilde{\mu},m_2]+$$

$$6C_{\phi2^2}g_2^2y_d^{i1i2}LF_{5,1,-3}[\tilde{\mu},m_2]-3m_2\tilde{\mu}g_2^2y_d^{i1i2}\left(\overline{C_{\phi2\phi1}}+C_{\phi2\phi1}\right)LF_{5,1,-2}[\tilde{\mu},m_2]-$$

$$\frac{1}{4}\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{2,1,-1}[\tilde{\mu},m_{\tilde{d}}^{r}]-\frac{1}{2}\overline{y_d}^{pr}y_d^{pi2}y_d^{i1r}LF_{2,1,-1}[\tilde{\mu},m_{\tilde{q}}^{p}]-$$

$$\frac{1}{4}y_d^{pi2}\overline{y_u}^{pr}y_u^{i1r}LF_{2,1,-1}[\tilde{\mu},m_{\tilde{u}}^{r}]+\frac{1}{9}m_1g_1^2a_d^{i1i2}LF_{1,1,1,0}[m_1,m_{\tilde{d}}^{i2},m_{\tilde{q}}^{i1}]-$$

$$\frac{1}{36}m_1g_1^2\left(C_{\phi2^2}a_d^{i1i2}+\overline{C_{\phi2\phi1}}\tilde{\mu}y_d^{i1i2}\right)LF_{2,2,1,-1}[m_1,m_{\tilde{d}}^{i2},m_{\tilde{q}}^{i1}]-$$

$$\frac{1}{3}g_1^2y_d^{i1i2}LF_{1,1,1,-1}[m_1,m_{\tilde{d}}^{i2},\tilde{\mu}]-\frac{1}{36}m_1g_1^2\left(C_{\phi2^2}a_d^{i1i2}+\overline{C_{\phi2\phi1}}\tilde{\mu}y_d^{i1i2}\right)$$

$$LF_{2,2,1,-1}[m_1,m_{\tilde{q}}^{i1},m_{\tilde{d}}^{i2}]-\frac{1}{6}g_1^2y_d^{i1i2}LF_{1,1,1,-1}[m_1,m_{\tilde{q}}^{i1},\tilde{\mu}]-$$

$$\frac{1}{6}C_{\phi2^2}g_1^2y_d^{i1i2}LF_{2,2,1,-2}[m_1,\tilde{\mu},m_{\tilde{d}}^{i2}]+\frac{1}{6}m_1\overline{C_{\phi2\phi1}}\tilde{\mu}g_1^2y_d^{i1i2}LF_{2,2,1,-1}[m_1,\tilde{\mu},m_{\tilde{d}}^{i2}]-$$

$$\frac{1}{12}C_{\phi2^2}g_1^2y_d^{i1i2}LF_{2,2,1,-2}[m_1,\tilde{\mu},m_{\tilde{q}}^{i1}]+\frac{1}{12}m_1\overline{C_{\phi2\phi1}}\tilde{\mu}g_1^2y_d^{i1i2}LF_{2,2,1,-1}[m_1,\tilde{\mu},m_{\tilde{q}}^{i1}]-$$

$$\frac{3}{2}g_2^2y_d^{i1i2}LF_{1,1,1,-1}[m_2,m_{\tilde{q}}^{i1},\tilde{\mu}]-\frac{3}{4}C_{\phi2^2}g_2^2y_d^{i1i2}LF_{2,2,1,-2}[m_2,\tilde{\mu},m_{\tilde{q}}^{i1}]+$$

$$\frac{3}{4}m_2\overline{C_{\phi2\phi1}}\tilde{\mu}g_2^2y_d^{i1i2}LF_{2,2,1,-1}[m_2,\tilde{\mu},m_{\tilde{q}}^{i1}]-\frac{8}{3}m_3g_3^2a_d^{i1i2}LF_{1,1,1,0}[m_3,m_{\tilde{d}}^{i2},m_{\tilde{q}}^{i1}]+$$

$$\frac{2}{3}m_3g_3^2\left(C_{\phi2^2}a_d^{i1i2}+\overline{C_{\phi2\phi1}}\tilde{\mu}y_d^{i1i2}\right)LF_{2,2,1,-1}[m_3,m_{\tilde{d}}^{i2},m_{\tilde{q}}^{i1}]+$$

$$\frac{2}{3}m_3g_3^2\left(C_{\phi2^2}a_d^{i1i2}+\overline{C_{\phi2\phi1}}\tilde{\mu}y_d^{i1i2}\right)LF_{2,2,1,-1}[m_3,m_{\tilde{q}}^{i1},m_{\tilde{d}}^{i2}]+$$

$$\frac{1}{18}m_1g_1^2\left(C_{\phi2^2}a_d^{i1i2}+\overline{C_{\phi2\phi1}}\tilde{\mu}y_d^{i1i2}\right)LF_{2,1,1,0}[m_{\tilde{d}}^{i2},m_1,m_{\tilde{q}}^{i1}]-$$

$$\frac{1}{18}m_1g_1^2\left(C_{\phi2^2}a_d^{i1i2}+\overline{C_{\phi2\phi1}}\tilde{\mu}y_d^{i1i2}\right)LF_{3,1,1,-1}[m_{\tilde{d}}^{i2},m_1,m_{\tilde{q}}^{i1}]-$$

$$\frac{4}{3}m_3g_3^2\left(C_{\phi2^2}a_d^{i1i2}+\overline{C_{\phi2\phi1}}\tilde{\mu}y_d^{i1i2}\right)LF_{2,1,1,0}[m_{\tilde{d}}^{i2},m_3,m_{\tilde{q}}^{i1}]+$$

$$\frac{4}{3}m_3g_3^2\left(C_{\phi2^2}a_d^{i1i2}+\overline{C_{\phi2\phi1}}\tilde{\mu}y_d^{i1i2}\right)LF_{3,1,1,-1}[m_{\tilde{d}}^{i2},m_3,m_{\tilde{q}}^{i1}]-$$

$$\tilde{\mu}^2y_d^{pi2}\overline{y_u}^{pr}y_u^{i1r}LF_{1,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},\tilde{\mu}]-\frac{1}{2}\tilde{\mu}y_d^{pi2}y_u^{i1r}\left(\overline{C_{\phi2\phi1}}\,\overline{a_u}^{pr}+C_{\phi2^2}\tilde{\mu}\overline{y_u}^{pr}\right)$$

$$LF_{2,1,1,0}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},\tilde{\mu}]+\frac{1}{2}\tilde{\mu}y_d^{pi2}y_u^{i1r}\left(\overline{C_{\phi2\phi1}}\,\overline{a_u}^{pr}+C_{\phi2^2}\tilde{\mu}\overline{y_u}^{pr}\right)LF_{3,1,1,-1}[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},\tilde{\mu}]+$$

$$\frac{1}{4}\tilde{\mu}y_d^{pi2}y_u^{i1r}\left(\overline{C_{\phi2\phi1}}\,\overline{a_u}^{pr}+C_{\phi2^2}\tilde{\mu}\overline{y_u}^{pr}\right)LF_{2,2,1,-1}[m_{\tilde{q}}^{p},\tilde{\mu},m_{\tilde{u}}^{r}]+$$

$$\frac{1}{18}m_1g_1^2\left(C_{\phi2^2}a_d^{i1i2}+\overline{C_{\phi2\phi1}}\tilde{\mu}y_d^{i1i2}\right)LF_{2,1,1,0}[m_{\tilde{q}}^{i1},m_1,m_{\tilde{d}}^{i2}]-$$

$$\frac{1}{18}m_1g_1^2\left(C_{\phi2^2}a_d^{i1i2}+\overline{C_{\phi2\phi1}}\tilde{\mu}y_d^{i1i2}\right)LF_{3,1,1,-1}[m_{\tilde{q}}^{i1},m_1,m_{\tilde{d}}^{i2}]-$$

$$\frac{4}{3}m_3g_3^2\left(C_{\phi2^2}a_d^{i1i2}+\overline{C_{\phi2\phi1}}\tilde{\mu}y_d^{i1i2}\right)LF_{2,1,1,0}[m_{\tilde{q}}^{i1},m_3,m_{\tilde{d}}^{i2}]+$$

$$\frac{4}{3}m_3g_3^2\left(C_{\phi2^2}a_d^{i1i2}+\overline{C_{\phi2\phi1}}\tilde{\mu}y_d^{i1i2}\right)LF_{3,1,1,-1}[m_{\tilde{q}}^{i1},m_3,m_{\tilde{d}}^{i2}]-$$

$$\frac{1}{2}\tilde{\mu}y_d^{pi2}y_u^{i1r}\left(\overline{C_{\phi2\phi1}}\,\overline{a_u}^{pr}+C_{\phi2^2}\tilde{\mu}\overline{y_u}^{pr}\right)LF_{2,1,1,0}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},\tilde{\mu}]+$$

$$\frac{1}{2}\tilde{\mu}y_d^{pi2}y_u^{i1r}\left(\overline{C_{\phi2\phi1}}\,\overline{a_u}^{pr}+C_{\phi2^2}\tilde{\mu}\overline{y_u}^{pr}\right)LF_{3,1,1,-1}[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},\tilde{\mu}]+$$

$$\left.\frac{1}{4}\tilde{\mu}y_d^{pi2}y_u^{i1r}\left(\overline{C_{\phi2\phi1}}\,\overline{a_u}^{pr}+C_{\phi2^2}\tilde{\mu}\overline{y_u}^{pr}\right)LF_{2,2,1,-1}[m_{\tilde{u}}^{r},\tilde{\mu},m_{\tilde{q}}^{p}]\right)$$